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Startup Product • 2024   Tech + Market Analysis   Product Design   Computer Vision

## Aqua 4.0 (ShrimpVision)

Mobile-first AI product to automate shrimp larvae counting and generate biomass + operational insights.

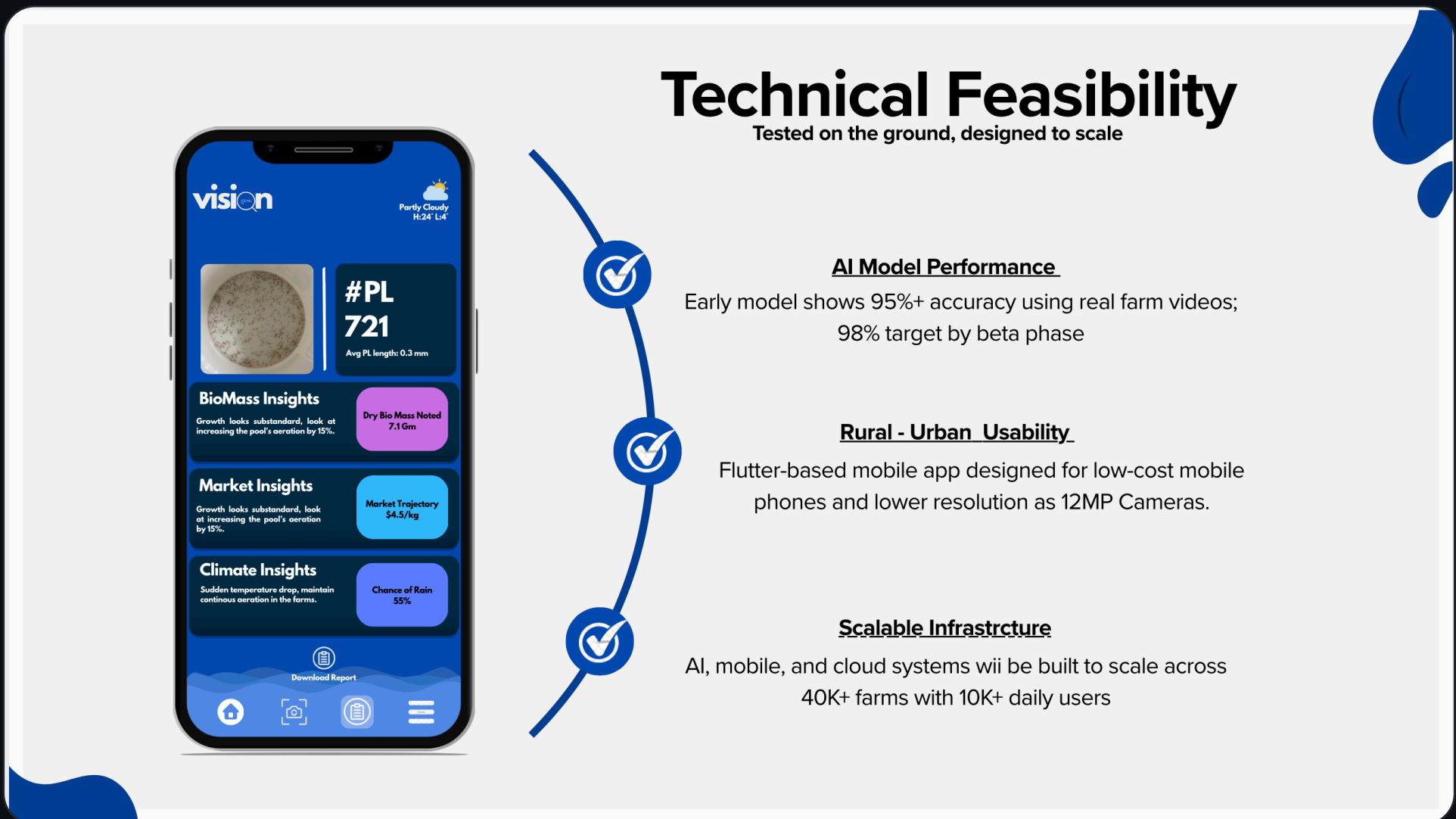
Role: CTO — led technical strategy, product architecture, model feasibility analysis, and mobile UX concept design.

Counting accuracy target  
~95% beta → goal 98%

Hardware requirement  
Smartphone camera only

Pilot scope  
12 hatcheries + 150–200 farms planned

Scale design  
40k+ farms, multi-region roadmap



### Problem

Shrimp larvae counting is still manual and inconsistent, leading to stocking errors, feed inefficiency, pricing disputes, and unreliable farm planning. Hatcheries need a fast, repeatable, and accessible way to measure counts and biomass without expensive equipment.

### Approach

- Defined the core product workflow: scan larvae → AI counts + sizes → biomass estimate → downloadable report + insights.
- Led technical feasibility analysis for smartphone-only deployment, model constraints, and image quality handling.
- Designed the product architecture including mobile UX flow, reporting logic, and CRM-style tracking of farm history.
- Benchmarked competitors and positioned the product around accessibility and scalability instead of hardware dependency.
- Worked with the team to define the pricing model, pilot rollout plan, and product roadmap.

### What I Owned

- Led technical feasibility analysis for smartphone-based counting
- Designed product workflow and mobile UX concept
- Defined system architecture and deployment approach
- Contributed to pricing, rollout, and go-to-market strategy

### Impact

- Transforms larvae counting into a repeatable digital workflow instead of manual estimation.
- Extends beyond counting into biomass insights and farm-level decision support.
- Creates a scalable mobile-first product viable for small and mid-size farms.
- Demonstrates end-to-end product thinking: problem framing, technical feasibility, UX design, pricing, and rollout.

### Deployment & Monitoring

- MVP architecture: mobile capture → preprocessing → model inference → cloud sync → report generation.
- Pilot rollout planned across hatcheries and farms with structured feedback collection.
- Product feedback loop via in-app signals, field agents, and usage analytics.
- Scale roadmap: India launch → Southeast Asia → Latin America → expansion to additional species.

### Risks & Tradeoffs

- Image quality variation (lighting, water clarity, device differences) requires robust preprocessing and model updates.
- Adoption risk if workflow is slower than manual methods — UX simplicity is critical.
- Connectivity constraints require offline-first capability and reliable sync behavior.

### Tools

Computer Vision   Mobile UX design   Product architecture planning   Google Cloud integration   Flutter & Firebase

### Deep Dive

Pitch Deck/ Mockup Design   Tech + Market Feasibility   Revenue Model